PROJECT ZEPHYRAQUUS
ENGINEERING NOTEBOOK
PROTOTYPE V0.4 – ESC MOUNTING & POWER CONNECTIONS

Objective-004:

Determine the final ESC mounting locations, power-distribution routing, and cable lengths prior to electrical assembly.

DG-004:

- **Mount both SpeedyBee F405 stacks.**
- **Verify motor-wire reach to each ESC.**
- **Route the XT60 power harness.**
- **Verify wiring accessibility for soldering.**
- **Maintain organized cable routing.**

DD-004:

1. **A 1-ft XT60 Y-splitter was selected in place of the original 100 mm splitter to provide sufficient cable reach and eliminate the connector mismatch without requiring additional extensions.**
2. **A small, rough cable pass-through was added near the center of the board to provide a more direct route from the XT60 Y-splitter to both ESCs while allowing the battery to remain positioned near the center of gravity.**
3. **The ESC power connections were physically mocked up before fabrication of the custom power cables to verify connector compatibility, cable routing, and required wire lengths prior to soldering. Only the lower ESC connection was represented in the mock-up because the same wiring configuration will be used for the upper ESC, and only one suitable wire was available and was left uncut until final cable lengths are determined.**

ECE-004:

- Maintaining consistent progress was difficult due to school and other responsibilities, resulting in an approximately one-month break from development.
- Battery placement was reconsidered because previous layouts placed it on the motor side, while the developing power-routing layout required access to the same central area.
- Limited space near the center of gravity complicated Y-splitter routing, leading to the addition of a rough cable pass-through.

Figure 4. V0.4 power-routing mock-up showing the cable pass-through, XT60 Y-splitter, and planned ESC connections prior to custom wire fabrication.